

CHATPULSE LITE

Mobile WhatsApp Activity Insights Dashboard

Fictional-Data Portfolio Prototype

Field	Details
Document	Complete Business Analysis, Delivery, Security and Hosting Evidence Pack
Document date	XX March 20XX
Prepared by	Shiv - Business Analyst
Project status	Public fictional-data prototype accepted with recorded limitations
Data classification	Fictional metadata only - no private communication content
Public deployment	https://chatpulse-lite.vercel.app/

Important product statement
ChatPulse Lite measures communication activity, not communication content. The current deployed version uses fictional sample data only and has no connection to personal WhatsApp accounts, chats, groups or media.

Professional Portfolio Edition

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All dates are intentionally anonymised as XX March 20XX for portfolio use.

No screenshots are included. This prevents cropped or distorted visual evidence and keeps the pack based on verified requirements, test results and release records.

1. Executive Summary

ChatPulse Lite: Mobile WhatsApp Activity Insights Dashboard is a mobile-focused activity dashboard prototype designed to present current unread and unseen communication activity using fictional metadata. The product was defined, designed, built, tested, accepted, prepared for serverless hosting, published to GitHub and deployed publicly on Vercel Hobby.

The business objective was to demonstrate how users could view activity counts by fictional person or group without storing or displaying private communication content. The dashboard presents unread message counts, unseen image counts, unseen video counts, people, groups, current activity timestamps and mobile-friendly navigation.

Count activity, never read content

One sent message event counts as one message regardless of how many words it contains. ChatPulse Lite is designed to count approved activity events and timestamps, not to read, analyse, display or retain the words, captions or media inside a communication.

Final delivery position

Area	Final outcome
Product definition and BA artefacts	Approved and completed
Mobile dashboard implementation	Completed using Next.js, TypeScript and Tailwind CSS
Fictional dataset	30 records: 20 people and 10 groups
Functional, regression and security verification	Passed with recorded limitations
Product Owner UAT	Approved with limitations
Vercel Hobby deployment	Completed
Physical Android live UAT	Passed
Current data status	Fictional metadata only; no real WhatsApp integration
Future real-data work	Not started; separate feasibility and security approval required

2. Project Overview and Chronological Delivery Timeline

Business problem

Users can find it difficult to understand how much unread or unseen activity exists across multiple people and groups. The project tested whether a simple mobile dashboard could summarise that activity while maintaining strict privacy boundaries and avoiding private content storage.

Chronological delivery record

Stage	Outcome	Date	Status
1. Initiation	Project purpose, product name, roles, fictional-data condition and privacy boundaries defined.	XX March 20XX	Approved
2. Requirements	Scope, business rules, user journeys, acceptance criteria and exclusions documented.	XX March 20XX	Approved
3. Data and risk design	Permitted metadata, prohibited fields, data dictionary, risk register and decision log prepared.	XX March 20XX	Approved
4. UX design	Mobile dashboard, chats list, detail view and navigation wireframes defined.	XX March 20XX	Approved
5. Local implementation	Next.js project, fictional data, API routes, dashboard, refresh and chat journeys developed.	XX March 20XX	Completed
6. Quality assurance	Data validation, API tests, regression tests, security checks and responsive testing completed.	XX March 20XX	Passed
7. UAT and local closure	Product Owner accepted the local fictional-data prototype with limitations; local Git checkpoint created.	XX March 20XX	Approved
8. Hosting assessment	Vercel Hobby selected; serverless filesystem risk identified and stateless refresh approved.	XX March 20XX	Approved
9. Hosting preparation	Runtime file writes removed; production CSP hydration defect diagnosed and corrected.	XX March 20XX	Completed
10. Public release	Repository published, Vercel deployment completed and public URL recorded.	XX March 20XX	Completed
11. Live mobile UAT	Physical Android detail-page refresh retest passed; suspected redirect defect not reproduced.	XX March 20XX	Passed
12. Portfolio closure	Professional delivery, security, traceability and hosting evidence consolidated.	XX March 20XX	Complete

Evidence references

Evidence	Reference
Local prototype Git checkpoint	29726df - Complete ChatPulse Lite fictional-data prototype v1.0.0
Hosting-readiness Git checkpoint	0a96d70 - Prepare ChatPulse Lite for Vercel Hobby hosting
Repository	https://github.com/singhsatyam85-cloud/chatpulse-lite
Public deployment	https://chatpulse-lite.vercel.app/

3. Business Requirements, Scope and Success Criteria

Business objective

Deliver a professional mobile activity dashboard prototype that displays current unread and unseen counts using fictional data, supports manual user-driven refresh and demonstrates privacy-first Business Analysis and software delivery practice.

Functional requirements

ID	Capability	Requirement
FR-01	Dashboard summary	Show total unread messages, unseen images, unseen videos, people and groups.
FR-02	Activity insights	Show leading fictional person/group activity and media summary information.
FR-03	Manual refresh	Refresh data on demand without a full-page reload and show loading, success or safe error states.
FR-04	Chats directory	Show all fictional people and groups with search, type filter and sorting controls.
FR-05	Detail view	Show current activity counters, chat type, timestamps and fictional participant details where applicable.
FR-06	Mobile navigation	Provide clear Dashboard and Chats navigation suitable for mobile use.
FR-07	Development simulation	Provide development-only fictional counter simulation and baseline reset controls.
FR-08	Production safety	Prevent development routes and local JSON files from being exposed publicly.
FR-09	Hosting compatibility	Operate without persistent server filesystem writes on serverless hosting.

Non-functional requirements

ID	Requirement	Measure
NFR-01	Mobile usability	No horizontal scrolling at tested mobile widths; interactive targets suitable for touch.
NFR-02	Privacy by design	No message text, captions, media or complete payloads stored or displayed.
NFR-03	Security	Secrets protected, server boundaries enforced, safe error handling and security headers enabled.
NFR-04	Reliability	Valid routes return expected responses and invalid chat IDs return genuine HTTP 404.
NFR-05	Maintainability	TypeScript, modular components, documented business rules and repeatable tests.
NFR-06	Cost control	No paid database, storage, API or hosting service introduced for the prototype.
NFR-07	Transparency	Public documentation must state fictional data and no real WhatsApp integration.

In scope

- Mobile-focused activity dashboard and chats directory.
- Current unread message, unseen image and unseen video counters.
- People and groups counts, search, filters, sorting and detail pages.
- Manual refresh and development-only simulation controls.
- Fictional local JSON baseline and request-stateless hosted refresh.
- Local and public testing, UAT, release documentation and portfolio evidence.

Out of scope

- Access to personal WhatsApp accounts, existing chats, family groups, friends groups or office groups.
- Real WhatsApp/Meta integration, QR login, unofficial scraping or personal chat export processing.
- Message content, captions, photos, videos, voice notes, documents, media IDs or complete webhook payloads.
- Historical snapshots, trend analytics, read-history tracking or automatic background refresh.
- Native Android APK or iOS package in the current version.
- Real customer authentication, multi-tenancy, paid production infrastructure or production support.

Success criteria

- Approved fictional dataset contains exactly 20 people and 10 groups.
- Dashboard and chat journeys work on mobile-sized screens.
- All core tests, lint and production build pass.
- Development-only routes remain unavailable in production.
- Public Vercel deployment loads successfully and live mobile refresh passes.
- No private communication content or real customer data is introduced.

4. User Stories and Acceptance Criteria

The following user stories replace internal delivery-task numbering with professional product-focused identifiers. Each story is linked to business acceptance criteria.

Story ID	Title	User story	Key acceptance criteria
CP-US-01	View activity overview	As a mobile dashboard user, I want to see current unread and unseen totals so that I can understand activity at a glance.	Dashboard shows messages, images, videos, people and groups; values are non-negative; loading and error states are clear.
CP-US-02	Find a person or group	As a mobile dashboard user, I want to search and filter chats so that I can locate a fictional person or group quickly.	Search is case-insensitive; People and Groups filters work; result count updates; empty state is understandable.
CP-US-03	Sort activity	As a mobile dashboard user, I want to sort chats by relevant activity measures so that I can prioritise review.	Approved sort options reorder results correctly and keep filters/search active.
CP-US-04	View chat details	As a mobile dashboard user, I want to open a person or group record so that I can view its current activity snapshot.	Valid ID loads; invalid ID returns HTTP 404; group participants remain fictional; no message content appears.
CP-US-05	Refresh on demand	As a mobile dashboard user, I want to refresh current activity manually so that I control when the visible snapshot changes.	Refresh prevents rapid duplicate actions; no full-page reload; success/error feedback shown; hosted response is not persisted.
CP-US-06	Use the dashboard on mobile	As a mobile user, I want clear touch navigation and readable layouts so that I can use the product comfortably on a phone.	Navigation remains available; no horizontal scroll at tested widths; controls meet practical touch-size expectations.
CP-US-07	Test fictional scenarios safely	As a tester, I want development-only simulation controls so that I can verify boundary conditions without using real data.	Controls work in development; baseline can be restored; development page/API return 404 in production.
CP-US-08	Protect private communication	As a privacy stakeholder, I want the system to store only approved counts and timestamps so that communication content is never exposed.	Prohibited fields absent; JSON files not public; errors/logs do not expose content, secrets or local paths.
CP-US-09	Demonstrate publicly without persistence	As a Product Owner, I want a free hosted demonstration so that portfolio reviewers can access the fictional prototype safely.	Vercel Hobby used; refresh is request-stateless; no database or cloud storage; public URL passes live UAT.

Acceptance principle

Activity event, not word count

A single sent item such as "aaa hjhd djks" contains three words but represents one message event. The unread message counter increases by one, not three. Multiple separately sent items are counted separately.

5. Data Model, Privacy Rules and Counting Logic

Permitted fictional metadata

Field	Purpose	Rule
id	Uniquely identify a fictional record.	Non-sensitive fictional identifier.
name	Display a fictional person or group name.	Fictional values only.
type	Distinguish Person and Group.	Controlled value.
unreadMessages	Current unread message-event count.	Non-negative integer.
unseenImages	Current unseen image-event count.	Non-negative integer.
unseenVideos	Current unseen video-event count.	Non-negative integer.
lastActive	Show latest fictional activity time.	Current snapshot only.
lastRefreshed	Show when the visible snapshot was generated.	Current request/snapshot only.
participants	Support fictional group-detail testing.	Fictional names/counts only.

Prohibited information

- Message words, message bodies, captions or semantic analysis results.
- Photos, videos, voice notes, documents, stickers, GIF files or downloaded media.
- Media identifiers, complete webhook payloads or unrestricted provider payload copies.
- Real phone numbers, email addresses, contact lists, customer identities or organisation records.
- Historical snapshots, previous values, trend histories or read-history records.
- API keys, tokens, secrets, credentials, local paths or private debug output.

Counting rules

Incoming activity example	Approved count	Content handling
"aaa hjhd djks" sent once	+1 unread message	Words are not counted individually and are not stored.
Ten words sent as one item	+1 unread message	One send event equals one message event.
Three separately sent text items	+3 unread messages	Each validated event counts once.
Emoji-only, sticker-only or GIF-only item	+1 unread message	Event counted without content analysis.
One unseen image	+1 unseen image	Image file and caption are not stored.
One unseen video	+1 unseen video	Video file and caption are not stored.
Text and emoji in the same sent item	+1 unread message	Single event is not double counted.

Privacy design statement

ChatPulse Lite measures communication activity, not communication content. It records permitted counts and timestamps only. The current version uses fictional data; a future authorised integration must discard prohibited content before storage or logging.

6. Solution Design and Technical Architecture

Product and implementation form

Dimension	Definition
Product form	Mobile WhatsApp activity insights dashboard prototype.
Current implementation form	Responsive Next.js application accessed through a mobile or desktop browser.
Current data source	Local read-only fictional JSON baseline.
Hosted runtime model	Request-stateless serverless API responses on Vercel Hobby.
Real-data status	No real WhatsApp connection or personal-account access.

Technology stack

Layer	Technology	Purpose
User interface	Next.js / React / TypeScript	Mobile dashboard, chats directory and detail views.
Styling	Tailwind CSS	Responsive layouts and mobile touch design.
API	Next.js route handlers	Provide fictional baseline and request-stateless refresh responses.
Data	Local JSON	Read-only fictional metadata baseline.
Testing	Node test scripts, ESLint and production build	Validate privacy, schema, API behaviour and code quality.
Version control	Git and GitHub	Local checkpoints and public portfolio repository.
Hosting	Vercel Hobby	No-cost public demonstration.

Current request flow

Step	Current prototype behaviour
1	User opens Dashboard, Chats or a valid chat-detail route.
2	Application requests a fresh copy of fictional baseline metadata.
3	Server returns current fictional counts and timestamps.
4	User may tap Refresh manually.
5	Refresh route calculates approved fictional changes for that response only.
6	React state updates without a full-page reload.
7	A later GET request returns the original baseline because no hosted state is persisted.

Architecture decisions

- No automatic refresh timer was implemented; user control remains explicit.
- No persistent cloud database or storage was introduced.
- No module-level mutable state or /tmp persistence was used for hosted refresh.
- Local development simulation remained environment-gated and unavailable in production.
- Security headers were configured centrally in next.config.ts.

7. Development and Delivery Summary

Delivery area	Implemented outcome
Project governance	Permanent instructions and approved project specification created before implementation.
Data foundation	Exactly 30 fictional records created and validated: 20 people and 10 groups.
API foundation	GET chats and POST refresh routes implemented with safe responses.
Dashboard	Current unread, image, video, person and group totals plus activity insights.
Manual refresh	Loading, success, safe error handling and rapid-click prevention.
Development test panel	Fictional increases, decreases, zero boundaries, record creation and baseline reset.
Chats directory	Search, People/Groups filters and five sort options.
Chat detail	Person/group details, participants, current metrics and refresh.
Mobile usability	Shared bottom navigation, responsive layouts and practical touch targets.
Security hardening	Server-only boundary, safe errors, security headers and protected local files/routes.
Defect correction	Invalid chat ID changed from client-only handling to genuine server-side HTTP 404.
Hosting readiness	Runtime writes removed and refresh changed to request-stateless behaviour.

Key delivery controls

- No task was marked complete without terminal output, tests, build evidence or Git status.
- No real customer information was introduced during development or testing.
- No forced dependency fix or breaking downgrade was applied.
- No remote push occurred until the hosting decision and repository publication were approved.
- No proposed code change was accepted when runtime evidence showed it was unnecessary.

8. Functional, Regression and Security Testing

Automated verification

Verification	Result	Evidence
Data validation	Pass	Exactly 30 records, 20 people, 10 groups; prohibited fields absent.
Security check	Pass	Seven security categories passed.
API tests	Pass	Baseline, refresh behaviour, validation and route responses verified.
Development panel tests	Pass	Boundary, clamping, simulation and reset journeys verified.
ESLint	Pass	Exit code 0; no errors or warnings.
Production build	Pass	Next.js 16.2.10; compilation and TypeScript errors 0.
Dependency audit	Pass with limitation	Two moderate PostCSS/Next.js findings remained; no forced fix used.

Production route verification

Route	Expected HTTP status	Observed
/	200	200
/chats	200	200
/chats/chat_001	200	200
/chats/invalid_chat_999	404	404
/dev-test	404	404
/api/dev-test	404	404
/data/chats.json	404	404
/data/chats-baseline.json	404	404

Responsive and security evidence

- Responsive behaviour checked at 320, 360/390, 430 and 768 pixel widths.
- No horizontal scrolling observed in the tested primary journeys.
- Security headers included CSP, X-Content-Type-Options, Referrer-Policy, X-Frame-Options and Permissions-Policy.
- No local Windows paths or stack traces appeared in public responses.
- No external analytics, trackers, remote images, paid APIs or real-data services were added.
- Development-only controls and JSON files returned 404 in production.

Defect record

Defect	Problem	Correction	Retest
CPL-DEF-001	Invalid chat detail ID returned HTTP 200 because not-found handling was client-only.	Moved ID validation into a server component and separated the client detail component.	Invalid ID returned genuine HTTP 404; regression passed.

9. User Acceptance Testing and Product Owner Sign-off

Local UAT journeys

Journey	Expected business outcome	Result
Open dashboard	Current fictional summary is understandable on mobile.	Pass
Refresh dashboard	Visible counters change without full-page reload.	Pass
Search and filter chats	User can find people and groups quickly.	Pass
Sort activity	User can prioritise fictional records by selected measure.	Pass
Open person detail	Current counters and timestamps display safely.	Pass
Open group detail	Current counters and fictional participants display safely.	Pass
Invalid chat route	Application returns a genuine not-found response.	Pass
Privacy acceptance	No message content, real contacts or private media appear.	Pass

Recorded Product Owner decision

Field	Recorded value
Product Owner	Shiv
Decision date	XX March 20XX
Decision	PRODUCT OWNER UAT APPROVED WITH LIMITATIONS
Local prototype status	Accepted for local fictional-data prototype closure

Accepted limitations at UAT

- No authentication, authorisation, tenant separation or real customer access.
- No production database, rate limiting, secure sessions or real-data privacy approval.
- Two moderate dependency findings remained documented.
- No real WhatsApp or personal-account integration.

10. Local Release, Git Checkpoint and Prototype Closure

Release item	Evidence
Version	v1.0.0 local fictional-data prototype
Local commit	29726df
Commit message	Complete ChatPulse Lite fictional-data prototype v1.0.0
Working tree after commit	Clean
Remote push at local closure	Not performed
Local status	Accepted as a local fictional-data prototype with recorded limitations

Release documentation completed

- Professional README with scope, setup, test commands and limitations.
- Release notes for v1.0.0.
- Prototype closure summary.
- Security and privacy verification report.
- Functional regression report and UAT report.
- Final production build report.

11. Hosting Assessment and Decision Record

Decision field	Approved outcome
Hosting option	Vercel Hobby
Business purpose	No-cost public portfolio demonstration of the fictional-data prototype
Repository	https://github.com/singhsatyam85-cloud/chatpulse-lite
Public deployment	https://chatpulse-lite.vercel.app/
Cost	No hosting cost selected
Data condition	Fictional sample metadata only
Persistent storage	None
Paid database/API/storage	Not introduced
Decision date	XX March 20XX
Hosting status	Complete with recorded limitations

Selection rationale

Vercel Hobby was selected because it supports the existing Next.js implementation, integrates with GitHub and provides a simple no-cost deployment route for a personal portfolio prototype. The decision did not approve real data, paid services or production customer use.

Approved hosting conditions

- Keep all data fictional and publicly safe.
- Retain manual Refresh but make it request-stateless.
- Do not store refresh changes in the server filesystem, a database, cookies or localStorage.
- Do not connect real WhatsApp accounts or personal communications.
- Restrict repository and Vercel application access to the project only.

12. Hosting Technical Changes and Production Defect Resolution

Serverless compatibility issue

The original local refresh workflow wrote updated fictional values to data/chats.json and data/status.json. Serverless runtime file changes are not reliable or persistent between requests, so the workflow was unsuitable for Vercel hosting.

Approved stateless change

Area	Before	After
Baseline read	Mutable current JSON state was read.	A fresh copy of chats-baseline.json is read for every request.
Refresh	Counts were updated and written to local JSON.	Approved fictional changes are calculated and returned directly.
Persistence	Later requests could observe local changes.	Later GET requests return the original baseline.
Runtime state	Local files changed.	No file writes, module-level mutable state, /tmp storage, cookies or localStorage.
Cloud services	Not applicable.	No database or cloud storage added.

Approved fictional refresh response

Record	Baseline	Refresh response
Meera Shah	5 messages, 1 image, 0 videos	8 messages, 2 images, 0 videos
Rohan Malhotra	15 messages, 2 images	11 messages, 1 image
Family Reunion 2026	0 messages, 0 images, 0 videos	8 messages, 2 images, 1 video

Production hydration defect

Item	Evidence
Observed symptom	Dashboard and Chats remained on grey loading skeletons in production.
Console evidence	Inline bootstrap script blocked by CSP script-src 'self'.
Root cause	Next.js hydration could not complete because required inline bootstrap scripts were blocked.
Rejected hypothesis	BottomNavigation.tsx was suspected initially but browser evidence disproved it.
Correction	Production CSP changed to script-src 'self' 'unsafe-inline'.
Control retained	No external script domains, wildcard scripts or production unsafe-eval were added.
Security decision	Reduced CSP strictness accepted only for the fictional prototype; review required before real data.

Hosting-readiness checkpoint

Field	Value
Commit	0a96d70
Commit message	Prepare ChatPulse Lite for Vercel Hobby hosting
Files changed	src/lib/snapshot-store.ts and next.config.ts
Git review	Diff check passed; no secrets, private data or unrelated files

13. Public Deployment and Live Mobile UAT

Deployment record

Field	Release evidence
Platform	Vercel Hobby
Repository	https://github.com/singhsatyam85-cloud/chatpulse-lite
Branch	main
Deployed commit	0a96d70
Public URL	https://chatpulse-lite.vercel.app/
Deployment result	Build and deployment passed
Deployment date	XX March 20XX

Stateless API evidence

Step	Observed result
Initial GET /api/chats	HTTP 200; 30 records, 20 people and 10 groups; baseline values returned.
POST /api/chats/refresh	HTTP 200; approved fictional changes and current request timestamp returned.
Immediate second GET	Original baseline values returned; POST changes were not stored.
Conclusion	Complete request isolation confirmed.

Browser UAT

Area	Observed result	Status
Dashboard	Loaded successfully after CSP correction; skeleton placeholders disappeared.	Pass
Chats page	30 chats, search, filters, sorting and fictional cards displayed.	Pass
Refresh totals	Messages 362 to 369; images 98 to 100; videos 13 to 14.	Pass
Record totals	People remained 20, groups 10 and total chats 30.	Pass
Page behaviour	No full-page reload during refresh.	Pass

Final physical Android UAT

Test field	Evidence
Device	Physical Android mobile device
Public site	https://chatpulse-lite.vercel.app/
Record tested	Meera Shah chat-detail page
Baseline	Unread messages 5; unseen images 1; unseen videos 0
After Refresh Chat Data	Unread messages 8; unseen images 2; unseen videos 0
Navigation	User remained on the Meera Shah detail page
State updates	Last Active and Last Refreshed updated; Data Status remained Ready
Page behaviour	No full-page reload and no automatic Dashboard redirect
Result and defect decision	PASS. The suspected redirect defect was not reproduced; no code change was required.

14. Requirements Traceability Matrix

Req.	Requirement	Implementation	Evidence	Status
FR-01	Dashboard totals	Dashboard cards and API fields	API, browser and mobile UAT	Pass
FR-02	Activity insights	Dashboard insight cards	Functional and UAT review	Pass
FR-03	Manual refresh	POST refresh route and React state	Success, failure, isolation and mobile tests	Pass
FR-04	Chats directory	/chats search, filters and sort	Functional and browser UAT	Pass
FR-05	Detail view	/chats/[id]	Valid, invalid, participant and refresh tests	Pass
FR-06	Mobile navigation	Shared bottom navigation	Responsive and physical mobile UAT	Pass
FR-07	Development simulation	/dev-test and /api/dev-test	Development tests; production 404	Pass
FR-08	Production safety	Headers, server-only boundary and route protection	Security check and HTTP route verification	Pass
FR-09	Hosting compatibility	Request-stateless snapshot store	GET-POST-GET isolation test	Pass
NFR-01	Mobile usability	Responsive layouts and touch controls	320-768px checks and Android UAT	Pass
NFR-02	Privacy by design	Prohibited-field rules and fictional data	Data validation and security scan	Pass
NFR-03	Security	Headers, safe errors and secret protection	Security checks 7/7	Pass with limitation
NFR-04	Reliability	Server-side 404 and production build	Regression and route verification	Pass
NFR-05	Maintainability	TypeScript, modular components and documents	Lint/build/document review	Pass
NFR-06	Cost control	Vercel Hobby and local JSON	Hosting decision review	Pass
NFR-07	Transparency	Fictional-data and no-integration statements	UI/document review	Pass

15. Live Environment Data Security Architecture

The current public site is suitable only for fictional portfolio data. A future version that processes real customer metadata would require a separate production architecture and formal security approval. The controls below are requirements, not claims about the current prototype.

Core security principle

Count activity, never read content

The production system should increment approved counters from validated message or media events without analysing the words or storing communication content. If an authorised source payload contains prohibited content, the processing layer must discard it immediately after extracting the minimum approved metadata.

Required production controls

Security layer	Required control
Data minimisation	Collect only approved name/type, counters and timestamps; reject all non-required fields.
Content exclusion	Never store or display message text, captions, photos, videos, voice notes, documents, media IDs or complete payloads.
Secure authentication	Use strong account verification, MFA/passkeys where appropriate and secure session management.
Role-based access	Grant only the minimum dashboard permissions required for each user role.
Tenant isolation	Separate every customer/organisation and enforce tenant checks on every data request.
Encryption	Use HTTPS in transit and encryption at rest for approved metadata and backups.
Secret management	Keep API keys, tokens and database secrets in secure server-side secret storage, never frontend code or Git.
Source verification	Verify webhook/API authenticity, schema, timestamp, replay protection and request size before processing.
Safe processing	Extract approved event type/count/timestamp and immediately discard prohibited payload content.
Safe logging	Log technical IDs, outcomes and authorised access only; never content, payloads, tokens or private contact data.
Rate limiting and WAF	Limit abusive requests, block suspicious traffic and alert on anomalous behaviour.
Retention and deletion	Apply approved short retention, automatic expiry and verified secure deletion.
Audit trail	Record authorised dashboard access and administrative actions without private communication content.
Monitoring and incident response	Detect misuse, rotate credentials, investigate incidents and follow notification obligations.
Security assurance	Complete code scanning, dependency review, access-control tests, penetration testing and remediation.
Privacy governance	Complete data-flow mapping, legal-basis assessment, DPIA/privacy review and formal go-live sign-off.

Mandatory real-data go-live gates

1. Approve a lawful and officially supported data source and document what events it provides.
2. Approve the minimum-data schema and demonstrate that prohibited content is discarded.
3. Implement authentication, MFA/passkeys where suitable, role controls and tenant isolation.
4. Confirm encryption, secret management, webhook/API authenticity and replay protection.
5. Implement rate limits, monitoring, safe logging, retention and deletion controls.
6. Complete privacy assessment, threat modelling and penetration testing.

7. Resolve unacceptable security findings and obtain Product Owner, privacy and security sign-off.

Go-live restriction: No real customer or communication data may be connected until every mandatory gate is evidenced and formally approved. The Vercel Hobby fictional prototype must not be repurposed as a real-data production service.

16. Future Android Access and Message-Event Counting Model

A future native Android application would separate user login from communication-event ingestion. Logging into the Android dashboard would confirm who may view the dashboard; it would not itself grant access to WhatsApp messages or personal chats. Real activity counting would require a separate, approved and officially supported business data source; personal WhatsApp chats remain outside the approved scope.

Recommended login approach

Stage	Recommended control	Purpose
First registration	Verified company email or mobile number with one-time code	Confirm the authorised account owner.
Ongoing sign-in	Passkey or secure password with MFA	Reduce account takeover risk.
Device convenience	Android biometric unlock after secure enrolment	Allow quick local re-entry without replacing server authentication.
Session control	Short-lived tokens, secure refresh, device/session revocation	Limit damage if a device or token is compromised.
Access decision	Role and tenant validation on every API request	Prevent unauthorised dashboard access.

How counting would work

Step	Future authorised processing behaviour
1. Authenticate dashboard user	Confirm the user may access a specific organisation/dashboard.
2. Receive authorised activity event	Use a formally approved business integration; do not scrape personal WhatsApp data.
3. Validate event	Verify source, signature, timestamp, schema and replay protection.
4. Classify event type	Message, image or video event only; no semantic reading of content.
5. Increment once	One validated send event increases the matching counter by one.
6. Discard prohibited content	Remove words, captions, media, complete payload and unnecessary identifiers before storage/logging.
7. Store approved metadata	Retain only authorised counters, type and timestamp under the correct tenant.
8. Display aggregated result	Show the mobile dashboard user only the permitted current activity information.

Counting examples

User action	Counter result	Reason
Sends "aaa hjhd djks" once	+1 message	One send event, regardless of three words.
Sends "Hello" and "How are you?" separately	+2 messages	Two separate send events.
Sends one emoji-only item	+1 message	One message event.
Sends one image with a caption	+1 image	Image event counted; caption not stored or counted separately.
Sends one video	+1 video	Video event counted; file is not stored.

17. Risk Register, Known Limitations and Recommendations

Key risks and mitigations

Risk	Severity	Impact	Mitigation	Status
Public fictional prototype mistaken for real integration	High	Viewers may assume personal WhatsApp access.	Clear fictional-data and no-integration statements.	Controlled
No authentication on public site	High for real use	Anyone with URL can view fictional dashboard.	Accept only for fictional portfolio; require auth before real data.	Open limitation
CSP unsafe-inline	Medium	Reduced script-policy strictness.	No wildcards/external scripts; replace with nonce/hash approach before real data.	Open limitation
Moderate dependency findings	Medium	Two PostCSS/Next.js findings remain.	Monitor and apply safe compatible update; no forced downgrade.	Open limitation
Metadata privacy	High	Counts and timestamps can still reveal behaviour.	Minimise, encrypt, restrict access, shorten retention and audit use.	Future control
Cross-tenant exposure	Critical	One customer could see another customer data.	Tenant isolation and authorisation on every query.	Future blocker
Payload/content leakage	Critical	Private words or media could reach storage/logs.	Schema allowlist, content discard, safe logging and tests.	Future blocker
Unofficial data collection	Critical	Legal, privacy and account risks.	Use only approved supported integration after feasibility review.	Future blocker
Duplicate event counting	Medium	Counters may be inaccurate.	Short-lived non-content deduplication/replay controls.	Future control
Excessive retention	High	Unnecessary privacy and breach exposure.	Approved retention schedule and secure deletion.	Future control

Current known limitations

- The deployed site uses fictional data and does not connect to WhatsApp or real customers.
- The current implementation is a responsive Next.js application, not a native Android APK.
- Public access has no authentication or tenant separation because the data is fictional.
- Hosted refresh is intentionally request-stateless and does not persist changes.
- Production CSP includes 'unsafe-inline' for the current Next.js hydration approach.
- Two moderate dependency findings remain monitored.
- No production database, real-data retention process, incident plan or privacy approval exists.

Recommended next steps

1. Keep the current public site fictional and stateless.
2. Update the professional PDF and portfolio description using the approved product name.
3. Monitor framework and dependency security updates.
4. Treat real-data feasibility as a separate project with formal discovery, legal/privacy review and supported API assessment.
5. Create a production security architecture before any Android or real-data implementation.
6. Do not use unofficial WhatsApp libraries, personal chat exports or device scraping.

18. Final Closure Statement and Delivery Checklist

Final project position

Closure decision

ChatPulse Lite: Mobile WhatsApp Activity Insights Dashboard is accepted as a public fictional-data portfolio prototype with recorded limitations. The local product, hosting preparation, deployment and physical Android live UAT are complete. Real customer data, personal WhatsApp access and production business use are not approved.

Closure item	Status
Project definition and permanent instructions approved	Complete
Business requirements, user stories and acceptance criteria documented	Complete
Privacy rules, data dictionary and risk decisions documented	Complete
Mobile application dashboard journeys implemented	Complete
Fictional dataset validated: 20 people and 10 groups	Complete
Functional, regression and security verification passed	Complete
Product Owner local UAT approved with limitations	Complete
Local Git checkpoint created	Complete
Vercel Hobby hosting assessment approved	Complete
Request-stateless hosting change implemented	Complete
Production CSP hydration issue corrected and retested	Complete
GitHub repository published	Complete
Vercel deployment completed	Complete
Public URL recorded	Complete
Physical Android live refresh UAT passed	Complete
Suspected redirect defect rejected after non-reproduction	Complete
Future real-data security requirements documented	Complete
Real-data feasibility and production go-live	Not started - separate approval required

Portfolio summary

This project demonstrates Business Analysis, privacy-by-design requirements, user stories, acceptance criteria, data modelling, risk management, Agile delivery support, functional testing, security verification, UAT, defect decision-making, Git release control, serverless hosting assessment and live mobile validation.

19. Appendices

Appendix A - Evidence reference

Reference	Value
Project name	ChatPulse Lite: Mobile WhatsApp Activity Insights Dashboard
Portfolio date	XX March 20XX
Local prototype commit	29726df
Hosting-readiness commit	0a96d70
Repository	https://github.com/singhsatyam85-cloud/chatpulse-lite
Public deployment	https://chatpulse-lite.vercel.app/
Local UAT decision	PRODUCT OWNER UAT APPROVED WITH LIMITATIONS
Live mobile UAT result	PASS
Final hosting status	Complete with recorded limitations

Appendix B - Delivery tracker completion comment

Jira-style template - Jira was not used in this project

ChatPulse Lite requirements, fictional-data implementation, testing, UAT, local release, hosting assessment, GitHub publication, Vercel deployment and physical Android live UAT are complete. The public prototype is available at <https://chatpulse-lite.vercel.app/>. Security checks, lint, production build and request-isolation testing passed. The suspected mobile redirect defect was not reproducible and no unnecessary code change was made. The product remains a fictional-data portfolio prototype with no real WhatsApp integration. Recorded limitations include no authentication, request-stateless refresh, two moderate dependency findings and the accepted CSP inline-script allowance.

Appendix C - Key commands and outcomes

Command	Outcome
node tests/validate-data.js	Pass - 30 records, 20 people, 10 groups
node tests/security-check.js	Pass - 7/7 categories
node tests/test-api.js	Pass
node tests/test-dev-panel.js	Pass
npm.cmd run lint	Pass - exit code 0
npm.cmd audit --omit=dev	Two moderate findings remained; no forced fix
npm.cmd run build	Pass - Next.js 16.2.10; zero TypeScript/compilation errors

Appendix D - Glossary

Term	Meaning
Activity event	One validated message, image or video occurrence counted once.
Request-stateless refresh	A response calculated for one request and not persisted for later requests.
Fictional metadata	Invented names, types, counts and timestamps used for development and demonstration.
UAT	User Acceptance Testing against approved business requirements.
CSP	Content Security Policy controlling browser resource execution.
Tenant isolation	Controls preventing one organisation from accessing another organisation data.